

INCLUDES

- ✓ Course framework
- ✓ Instructional section
- ✓ Sample exam questions

AP[®] Macroeconomics

COURSE AND EXAM DESCRIPTION

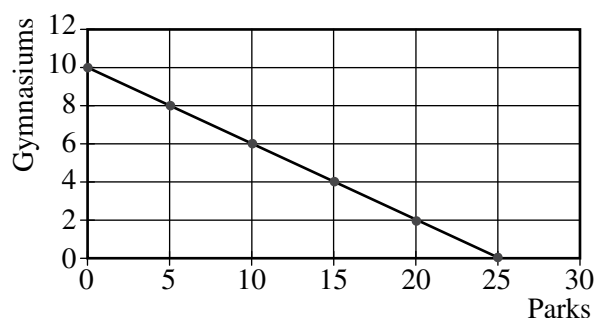
Effective
Fall 2022

Sample Exam Questions

The sample exam questions that follow illustrate the relationship between the course framework and the AP Macroeconomics Exam and serve as examples of the types of questions that appear on the exam. After the sample questions are tables that show which skill, learning objective(s), and unit each question relates to. The answers to the multiple-choice questions are also provided.

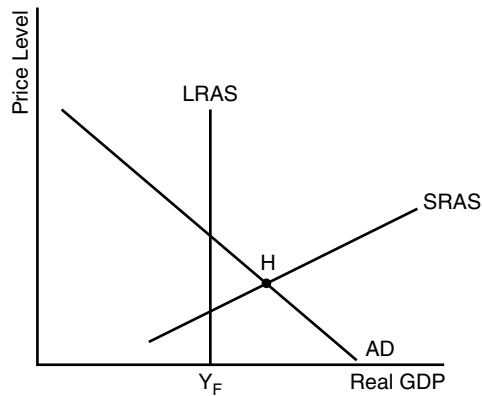
Section I: Multiple-Choice

The following are examples of the kinds of multiple-choice questions found on the exam.



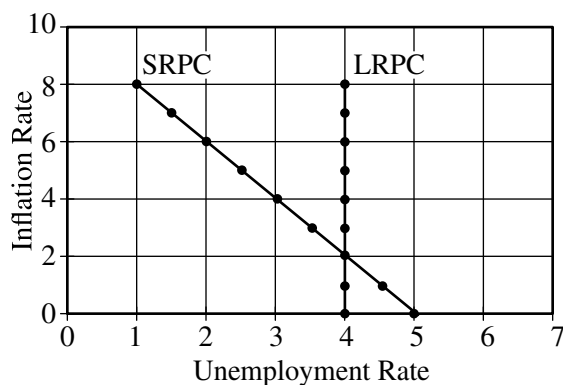
1. The graph above shows the production possibilities curve for a small township that is deciding to build parks and gymsnasiums. Which of the following combinations of parks and gymsnasiums is unattainable given the township's available resources?
(A) 5 parks and 6 gymsnasiums
(B) 5 parks and 8 gymsnasiums
(C) 10 parks and 6 gymsnasiums
(D) 15 parks and 4 gymsnasiums
(E) 20 parks and 4 gymsnasiums
2. Which of the following changes would result in an indeterminate change in the equilibrium price in a perfectly competitive market?
(A) An increase in demand and a decrease in supply
(B) An increase in demand and an increase in supply
(C) A decrease in demand and an increase in supply
(D) A decrease in demand with no change in supply
(E) A decrease in supply with no change in demand

3. Which of the following is accounted for in the calculation of a country's gross domestic product?
- (A) Sales of stocks and bonds
 - (B) Changes in inventories
 - (C) Changes in product quality
 - (D) The underground economy
 - (E) Nonmarket activities
4. The population of Country X is 250,000, and the labor force is 200,000 people. If 175,000 people are employed, what is the unemployment rate?
- (A) 10%
 - (B) 12.5%
 - (C) 20%
 - (D) 80%
 - (E) 87.5%
5. Which of the following will happen if the government raises both taxes and spending by \$100 million and the marginal propensity to consume is 0.8?
- (A) Real GDP will decrease by a maximum of \$500.
 - (B) Real GDP will decrease by a maximum of \$400.
 - (C) Real GDP will increase by a maximum of \$100.
 - (D) Real GDP will increase by a maximum of \$400.
 - (E) Real GDP will increase by a maximum of \$500.
6. Which of the following would lead to an increase in nominal interest rates?
- (A) An expansionary monetary policy accompanied by an increase in the demand for money
 - (B) An expansionary monetary policy accompanied by a decrease in the demand for money
 - (C) An expansionary monetary policy conducted without any change in the demand for money
 - (D) A contractionary monetary policy accompanied by an increase in the demand for money
 - (E) A contractionary monetary policy accompanied by a decrease in the demand for money



7. A country's economy is in equilibrium at point H. Which of the following policies would be most effective to reduce the price level in the short run?
 - (A) Increasing the minimum wage
 - (B) Increasing government expenditures
 - (C) Increasing interest on reserves
 - (D) Decreasing the required reserve ratio
 - (E) Decreasing income tax rates
8. Country H's current domestic output is lower than its potential domestic output. Assume that the central bank now decreases its administered interest rates. What will be the short-run effects of the central bank's action on cyclical unemployment and real income?
 - (A) Cyclical unemployment will increase, and real income will increase.
 - (B) Cyclical unemployment will increase, and real income will decrease.
 - (C) Cyclical unemployment will remain the same, and real income will increase.
 - (D) Cyclical unemployment will remain the same, and real income will decrease.
 - (E) Cyclical unemployment will decrease, and real income will increase.
9. An increase in the demand for loanable funds could be best explained by which of the following?
 - (A) A decrease in investment spending.
 - (B) An increase in the government's budget surplus.
 - (C) An increase in firms' optimism about the future performance of the country's economy.
 - (D) Domestic investors seeking higher returns by investing in foreign financial assets.
 - (E) An increase in political instability in the country.

10. If both contractionary monetary policy and contractionary fiscal policy are carried out, what will most likely happen to interest rates and real gross domestic product (GDP) in the short run?
- (A) Both interest rates and real GDP will increase.
 - (B) Both interest rates and real GDP will decrease.
 - (C) Interest rates will decrease, and real GDP will stay the same.
 - (D) Interest rates will increase, and real GDP will decrease.
 - (E) Real GDP will decrease, and the change in interest rates will be indeterminate.
11. If an economy is currently in a recessionary gap, which of the following changes would result in an increase in real GDP in the short run and a decrease in the price level in the long run?
- (A) The government begins running a budget surplus.
 - (B) There is an increase in real interest rates.
 - (C) The government increases income tax rates.
 - (D) There is an increase in the prices of the economy's productive resources.
 - (E) There is an increase in the productivity of the economy's resources.
12. If the government offers a tax credit to businesses, what will be the most likely effects of this action?
- (A) A decrease in consumption spending, an increase in aggregate demand, and an increase in real output
 - (B) An increase in consumption spending, a decrease in aggregate demand, and a decrease in real output
 - (C) An increase in investment spending, an increase in the capital stock, and an increase in real output
 - (D) A decrease in investment spending, a decrease in the capital stock, and an increase in real output
 - (E) A decrease in government spending, a decrease in aggregate demand, and a decrease in real output
13. If an economy experiences an improvement in technology, what will happen to its production possibilities curve (PPC) and its long-run aggregate supply (LRAS) curve?
- (A) Both curves shift inward.
 - (B) Both curves shift outward.
 - (C) The PPC shifts inward, and the LRAS curve stays the same.
 - (D) The PPC shifts outward, and the LRAS curve shifts inward.
 - (E) The PPC stays the same, and the LRAS curve shifts outward.



14. The diagram above shows the short-run Phillips curve (SRPC) and the long-run Phillips curve (LRPC) for an economy. If the rate of inflation is currently 6%, what is the current unemployment rate?
- (A) The current unemployment rate is 1%.
- (B) The current unemployment rate is 2%.
- (C) The current unemployment rate is 3%.
- (D) The current unemployment rate is 4%.
- (E) The current unemployment rate is 5%.
15. If the United States budget deficit increases, what will most likely happen to the United States dollar in the foreign exchange market?
- (A) It will appreciate because interest rates will increase.
- (B) It will depreciate because interest rates will decrease.
- (C) It will appreciate because interest rates will decrease.
- (D) It will depreciate because interest rates will increase.
- (E) It will not change because changes in the government budget have no effect on the exchange rate.

Section II: Free-Response

The following are examples of the kinds of free-response questions found on the exam. Note that on the actual AP Exam, there will be one long free-response question worth 10 points and two short free-response questions, each worth 5 points.

1. Assume that the country of Zeetoland is in short-run equilibrium. The expected inflation rate is 4 percent, the actual rate of unemployment is 5 percent, the natural rate of unemployment is 6 percent, and the equilibrium real interest rate is 3 percent.
 - (a) Draw a correctly labeled graph of the aggregate demand, short-run aggregate supply, and long-run aggregate supply curves for Zeetoland, and show each of the following.
 - (i) The current equilibrium real output and price level, labeled Y_1 and PL_1 respectively
 - (ii) The full-employment output, labeled Y_F

- (b) Assume no policy action is taken to address the output gap in Zeetoland. Explain how the economy will adjust to full employment in the long run.
- (c) Assume instead that the central bank is concerned about the buildup of inflationary pressures in Zeetoland and is considering taking action to fight inflation. Assuming the banking system in Zeetoland has ample reserves, identify one monetary policy action the central bank would likely take.
- (d) How would the monetary policy action identified in part (c) affect each of the following in the short run?
 - (i) The price of previously issued bonds in Zeetoland
 - (ii) Real output in Zeetoland. Explain.
 - (iii) Net financial capital flows to Zeetoland. Explain.
- (e) Draw a correctly labeled graph of the foreign exchange market for Zeetoland's currency, the zeet, and show the effect of the monetary policy action identified in part (c) on the demand and the exchange rate for the zeet.
- (f) Based on the change in the value of the zeet shown in part (e), what will happen to Zeetoland's exports? Explain.

2. The table shows macroeconomic statistics for the country of Fehran.

Year	Nominal GDP	Tax Revenues	Government Outlays	Consumption Spending	Population	GDP Deflator
2011	\$150,000	\$25,000	\$25,000	\$75,000	100	100
2012	\$225,000	\$30,000	\$35,000	\$100,000	120	125

- (a) Using the data in the table, calculate each of the following and show your work.
 - (i) Fehran's real GDP in 2012
 - (ii) The inflation rate in Fehran from 2011 to 2012
- (b) Fehran is the only producer and exporter of electric vehicles. Assume the price level among Fehran's trading partners remains the same and the inflation rate calculated in part (a)(ii) is reflective of price changes for electric vehicles. As a result of the change in the relative price level, would foreign importers buy more or fewer electric vehicles from Fehran in 2012 compared to 2011? Explain.
- (c) Did the standard of living of the average person in Fehran increase, decrease, or stay the same between 2011 and 2012? Explain.
- (d) What effect will the changes in Fehran's government budget between 2011 and 2012 have on Fehran's national debt? Explain.

Answer Key and Question Alignment to Course Framework

Multiple-Choice Question	Answer	Skill	Learning Objective	Unit
1	E	2.C	MOD-1.B	1
2	B	2.B	MKT-2.G	1
3	B	1.A	MEA-1.A	2
4	B	1.C	MEA-1.C	2
5	C	3.C	MOD-2.B, POL-1.A	3
6	D	2.B	POL-1.D, MKT-3.D	4
7	C	2.A	POL-1.D	4
8	E	3.A	POL-1.D	4
9	C	2.A	MKT-4.E	4
10	E	3.A	POL-1.F	5
11	E	2.A	MEA-2.B	5
12	C	3.A	POL-4.A	5
13	B	3.A	MOD-1.C	5
14	B	2.C	MOD-3.A	5
15	A	3.B	MKT-4.E, MKT-5.E	6

Free-Response Question	Question Type	Learning Objectives	Unit
1	Long	MOD-2.G, MOD-2.I, POL-1.D, MEA-3.A, MKT-5.G, MKT-5.C, MKT-5.E, MKT-5.F	3, 4, 6
2	Short	MEA-1.J, MEA-1.F, MOD-2.A, MEA-1.I, POL-3.B	2, 3, 5

The scoring information for the questions within this course and exam description, along with further exam resources, can be found on the [AP Macroeconomics Exam Page](#) on AP Central.



Scoring Guidelines

Question 1: Long

1. Assume that the country of Zeetoland is in short-run equilibrium. The expected inflation rate is 4 percent, the actual rate of unemployment is 5 percent, the natural rate of unemployment is 6 percent, and the equilibrium real interest rate is 3 percent.
 - (a) Draw a correctly labeled graph of the aggregate demand, short-run aggregate supply, and long-run aggregate supply curves for Zeetoland, and show each of the following.
 - (i) The current equilibrium real output and price level, labeled Y_1 and PL_1 respectively
 - (ii) The full-employment output, labeled Y_F
 - (b) Assume no policy action is taken to address the output gap in Zeetoland. Explain how the economy will adjust to full employment in the long run.
 - (c) Assume instead that the central bank is concerned about the buildup of inflationary pressures in Zeetoland and is considering taking action to fight inflation. Assuming the banking system in Zeetoland has ample reserves, identify one monetary policy action the central bank would likely take.
 - (d) How would the monetary policy action identified in part (c) affect each of the following in the short run?
 - (i) The price of previously issued bonds in Zeetoland
 - (ii) Real output in Zeetoland. Explain.
 - (iii) Net financial capital flows to Zeetoland. Explain.
 - (e) Draw a correctly labeled graph of the foreign exchange market for Zeetoland's currency, the zeet, and show the effect of the monetary policy action identified in part (c) on the demand and the exchange rate for the zeet.
 - (f) Based on the change in the value of the zeet shown in part (e), what will happen to Zeetoland's exports? Explain.

Scoring Guidelines for Question 1: Long

10 points

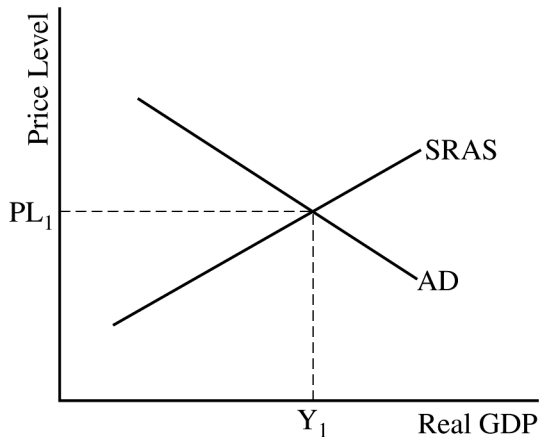
Learning Objectives: MOD-2.G MOD-2.I POL-1.D MEA-3.A MKT-5.G MKT-5.C MKT-5.E MKT-5.F

- (a) Draw a correctly labeled aggregate demand-aggregate supply graph that shows PL_1 and Y_1 at the intersection of AD and SRAS.

1 point

4.A

Task type: Create graphs or visual representations

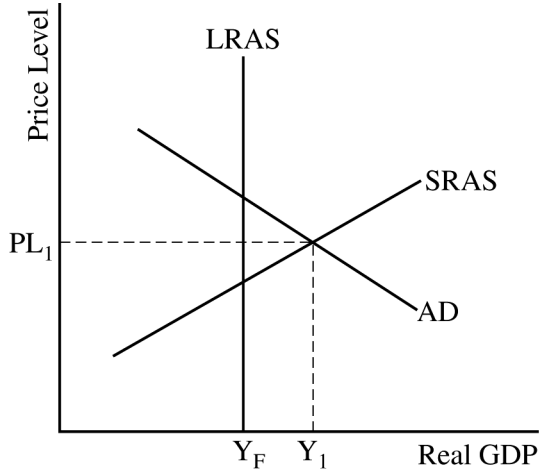


For the second point, the graph must show a vertical LRAS curve to the left of Y_1 and label the full employment output Y_F .

1 point

4.B

Task type: Create graphs or visual representations



Total for part (a)

2 points

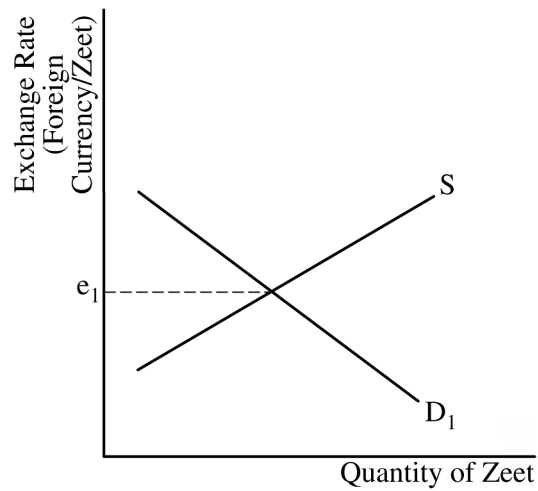
(b)	Explain that short-run aggregate supply will decrease in the long run as a result of an increase in nominal wages, input prices, and/or inflationary expectations. <i>Task type: Explain</i>	1 point 3.A
(c)	State that the central bank would increase its administered interest rates (e.g., increase interest on reserves). <i>Task type: Make assertions</i>	1 point 2.A
(d)	(i) State that the price of previously issued bonds in Zeetoland would decrease. <i>Task type: Make assertions</i>	1 point 3.A
	(ii) State that real output in Zeetoland would decrease and explain that the increase in nominal interest rates in Zeetoland would decrease interest-sensitive consumption and investment spending and decrease aggregate demand. <i>Task type: Explain</i>	1 point 3.A
	(iii) State that Zeetoland will experience net financial capital inflows and explain that higher interest rates in Zeetoland will attract financial capital from other countries. <i>Task type: Explain</i>	1 point 3.B
Total for part (d)		3 points

- (e) Draw a correctly labeled graph of the foreign exchange market for the zeet.

1 point

Task type: Create graphs or visual representations

4.A

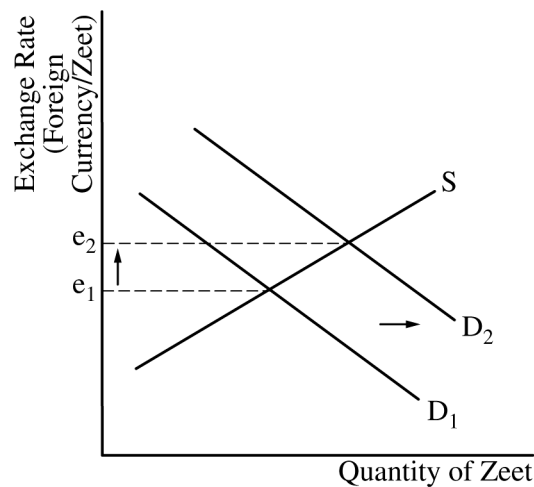


For the second point, the graph must show a rightward shift in the demand curve for the zeet and an increase in the exchange rate (an appreciation in the currency).

1 point

Task type: Create graphs or visual representations

4.C



Total for part (e)

2 points

- (f) State that Zeetoland's exports will decrease and explain that the appreciation of the zeet will make Zeetoland's goods and services relatively more expensive than other countries' goods and services.

1 point

Task type: Explain

3.A

Total for question 1

10 points

Question 2: Short

2. The table shows macroeconomic statistics for the country of Fehran.

Year	Nominal GDP	Tax Revenues	Government Outlays	Consumption Spending	Population	GDP Deflator
2011	\$150,000	\$25,000	\$25,000	\$75,000	100	100
2012	\$225,000	\$30,000	\$35,000	\$100,000	120	125

- (a) Using the data in the table, calculate each of the following and show your work.
- (i) Fehran's real GDP in 2012
 - (ii) The inflation rate in Fehran from 2011 to 2012
- (b) Fehran is the only producer and exporter of electric vehicles. Assume the price level among Fehran's trading partners remains the same and the inflation rate calculated in part (a)(ii) is reflective of price changes for electric vehicles. As a result of the change in the relative price level, would foreign importers buy more or fewer electric vehicles from Fehran in 2012 compared to 2011? Explain.
- (c) Did the standard of living of the average person in Fehran increase, decrease, or stay the same between 2011 and 2012? Explain.
- (d) What effect will the changes in Fehran's government budget between 2011 and 2012 have on Fehran's national debt? Explain.

Scoring Guidelines for Question 2: Short

5 points

Learning Objectives: MEA-1.J MEA-1.F MOD-2.A MEA-1.I POL-3.B

- (a) (i) Calculate Fehran's real GDP in 2012 as \$180,000 and show your work. **1 point**
 $(\$225,000/125) \times 100 = \$180,000$
1.C

Task type: Perform numerical analysis

- (ii) Calculate the inflation rate in Fehran from 2011 to 2012 as 25% and show your work. **1 point**
 $[(125-100) / 100] \times 100 = 25\%$
1.C

Task type: Perform numerical analysis

Total for part (a) 2 points

- (b) State that foreign importers of electric vehicles will buy fewer electric vehicles from Fehran in 2012 and explain that electric vehicles in Fehran are relatively more expensive as a result of the increase in the price level. **1 point**
2.C

Task type: Explain

- (c) State that the standard of living of the average person in Fehran stayed the same and explain that real GDP per capita did not change from 2011 to 2012 or that it stayed at \$1,500. **1 point**
2.C

Task type: Explain

- (d) State that Fehran's national debt will increase and explain that government outlays are greater than taxes and a government adds to the national debt when it runs a budget deficit. **1 point**
3.C

Task type: Explain

Total for question 2 5 points